

GW170817 glitch robustness comparison

Samples per model: 256

Custom: /Users/jayantkumar/Desktop/ADAPT-Project/checkpoints/glitch_detector_v1/best_glitch

Injected glitch (H1 only):

type: sine-Gaussian (blip-like)

t_rel: -1.0 s (relative to trigger)

f0: 100.0 Hz

Q: 5.0

amplitude: 3.728e-19 (8.0x RMS)

poison (glitch+Welch).

detect-and-gate + orig ASD.

Glitchy GW170817 — medians & 90% CI
poison (glitch+Welch) N=256 | detect-and-gate + orig ASD N=256

parameter	baseline median	baseline 90% CI	custom median	custom 90% CI
chirp_mass	1.19763	[1.19751, 1.19774]	1.19759	[1.19745, 1.1977]
mass_ratio	0.883475	[0.744351, 0.984365]	0.878474	[0.741682, 0.989342]
luminosity_distance	10.0663	[9.96412, 10.6306]	30.9342	[15.6516, 44.4623]
theta_jn	1.76644	[1.7409, 1.79079]	2.2866	[1.90681, 2.88253]

GW170817 glitch robustness: baseline vs custom

poison (glitch+Welch)
detect-and-gate + orig ASD

